Mohammed Raaed Al-Yousif

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AI & Robotics Engineering student at AUIS with applied experience in machine learning, neural network design, and full-stack software development. Research focused on predictive maintenance in rotating machinery and ANN-based solar irradiance prediction, both directly applicable to industrial and energy-sector challenges. Builder of three live, deployed web tools. Certified Document Control Specialist. Targeting opportunities in the Oil & Gas sector where AI and data-driven engineering can create measurable operational value.

EDUCATION

American University of Iraq, Sulaimani (AUIS) *2023 – 2027*

B.Sc. in Artificial Intelligence and Robotics Engineering

Relevant Coursework:
Machine Learning · Large Language Models · Algorithms & Data Structures · Mechatronics · Engineering Statistics · Introduction to AI · Circuits · Engineering Project Management · Research Writing · Computer Vision (in progress) · Senior Design I (in progress)

TECHNICAL SKILLS

Machine Learning & AI
TensorFlow · PyTorch · Scikit-learn · Neural Networks (FFNN, LSTM, CNN, RNN) · RAG Systems · LLMs · Computer Vision

Programming
Python · TypeScript · JavaScript · MATLAB

Data & Analytics
Data Preprocessing · Cross-Validation · Statistical Analysis · Minitab · Excel

Web & Software Development
React · Vite · Convex · Clerk · Tailwind CSS · Gradio · ChromaDB · Git/GitHub

Engineering Tools
AutoCAD · Engineering Drawing · Microsoft Suite

Industry Skills
Document Control (CDCS Certified) · Project Management · Photography (Sony mirrorless)

PROJECTS

GHI Prediction using Artificial Neural Networks *2025 – 2026*

Python · TensorFlow · Scikit-learn · GroupKFold Cross-Validation · Satellite Meteorological Data

- Designed and trained four ANN architectures (FFNN, CFNN, LSTM, EMNN) to predict Global Horizontal Irradiance (GHI) across 8 locations in Oman using satellite-derived climatological data.
- Applied feature engineering including cyclical time encoding, lag features, and spatial coordinates; produced publication-quality benchmarking reports at 300 DPI.
- Directly underpins an ongoing student research publication on solar irradiance forecasting.

Predictive Maintenance in Rotating Machinery *2026 (In Progress)*

Capstone Research Project — AUIS Senior Design I | Signal Processing · Fault Detection · Literature Review

- Leading the research track for a team capstone investigating AI-driven fault detection in rotating machinery — directly applicable to pumps, compressors, and turbines used in Oil & Gas operations.

– Conducted a systematic literature review of signal processing algorithms for fault detection, covering vibration analysis, FFT, and ML-based classification approaches across 18 academic papers.

Gradify 2.0 — GPA Calculator | Live at gradify.netlify.app *2025 – Present*

React · TypeScript · Convex (Serverless Backend) · Clerk Authentication · Tailwind CSS · Netlify

– Built and deployed a full-stack GPA management tool actively used by AUIS students; supports PDF transcript import, real-time GPA simulation, retake detection, and cross-device sync.
– Architected a serverless backend using Convex with authenticated data persistence per user; deployed continuously via Netlify with a separate Convex Cloud backend.

AUIS Academic Catalog RAG Assistant *2025 – 2026*

Python · Phi-3-mini (4-bit quantized) · ChromaDB · Sentence Transformers · Gradio · CUDA

– Built a fully local Retrieval-Augmented Generation (RAG) system that allows students to query the AUIS Academic Catalog in natural language — no cloud APIs, runs entirely on-device.
– Implemented embedding-based vector retrieval (ChromaDB + MiniLM), chunked PDF ingestion, and GPU-accelerated inference via 4-bit quantized Phi-3; deployed with a Gradio chat interface.

Emotion-Based Music Player *2025 – 2026*

Python · PyTorch · ResNet-18 · OpenCV · FER2013 Dataset · Real-time Inference

– Trained a ResNet-18 convolutional neural network from scratch on the FER2013 dataset to classify facial emotions into 4 categories with real-time webcam inference.
– Integrated emotion recognition with an adaptive music playback system; model uses a 15-prediction smoothing window to determine dominant emotion before triggering playlist changes.

Gamut — Interactive Photo Analysis Lab *2025 – Present*

React · TypeScript · WebGL · Vite · Netlify

– Developed a browser-based image processing tool with WebGL-accelerated real-time scopes (histogram, waveform, vectorscope) for photographic analysis.

RESEARCH ACTIVITY

Solar Irradiation Prediction — Student Publication *2026 (In Progress)*

ANN · Climatological Data · Oman Satellite Dataset

– Co-authoring a student research paper on ANN-based GHI prediction using satellite-derived meteorological data; manuscript drafting pending instructor feedback.

Predictive Maintenance Literature Survey *2025 – 2026*

Rotating Machinery · Fault Detection Algorithms · Signal Processing

– Completed a structured literature review for the capstone research track, synthesizing findings from 18 peer-reviewed papers on vibration-based and ML-based fault detection methods.

WORK EXPERIENCE

Video Editor & Filmmaker · Shift Agency *January 2025*

Internship

– Produced and edited video content for agency clients; managed end-to-end production workflow including filming, editing, and delivery using DaVinci Resolve.

CERTIFICATIONS & AWARDS

- Certified Document Control Specialist (CDCS)
- Foundations of Project Management
- Project Initiation: Starting a Successful Project
- AUIS Voluntary Service Award
- VEX Robotics Competition — Head Scorekeeper Referee

VOLUNTEERING

VEX Robotics Competition · AUIS Campus *2024*

- Served as Head Scorekeeper Referee — responsible for tallying team scores and reporting results to tournament officials across all match rounds.

AUIS Admissions Office *2023 – 2024*

- Assisted with admissions inquiries and compiled student registration data; accumulated 51–75 verified volunteer hours (Spring 2024 Award).

AUIS Podcast *2023 – 2024*

- Filmed and edited podcast episode recordings using DaVinci Resolve.

LANGUAGES

Arabic: Native **English**: Professional Proficiency